

Retirement Alpha and Beta

$$\alpha = \frac{2 \times AM + 2.773 / MRL}{VOL^2 + 0.6931 / MRL} - 1$$

$$\beta = \frac{2 \times S}{VOL^2 + 0.6931 / MRL}$$

Figure 8.3 The only equations you need for sustainable spending.

Source: M. Milevsky, "Sustainability and Ruin," *Research Magazine*, June 2007.

Finally, use the calculated retirement alpha and retirement beta variables and look up the relevant number under the row and column from Table 8.3 to arrive at the retirement sustainability probability. Remember, there are four inputs that you need to compute or calculate the retirement alpha and beta variables. The spending rate is denoted by the letter *S*. The average return you expect to earn from your investment portfolio is denoted by the letters *AM*, which stands for arithmetic mean. The average amount of time you spend in retirement is denoted by the letters *MRL*, which stands for median remaining lifetime, and the final variable is the investment volatility, denoted by the abbreviation *VOL*.

So, once again, start with *four* numbers. Convert them into *two* numbers. Then reduce that into *one* number, which is the sustainability of your strategy.

I know this might appear confusing at first, so here are some examples that convert four numbers into two and then into one. Let's imagine that I decide to retire today at the age of 40, when my median remaining lifespan (*MRL*) is exactly 40 years. Remember that this implies a 50% chance I'll get to age 80. Now assume that I invest whatever nest-egg I have today into a portfolio that is expected to earn an arithmetic average (*AM*) of 8% after inflation in any given year and that the volatility (*VOL*) of my portfolio's return is 20%. These numbers—which are on the optimistic side—can usually be estimated from the historical performance of my investments and my current asset allocation. Finally, I desire an annual spending rate of 5% adjusted for inflation. Remember, this means that if I start with \$1,000,000, then I'm spending \$50,000 per year, and if I start with \$100,000, then I'm spending \$5,000 per year. Both figures are adjusted annually for inflation.